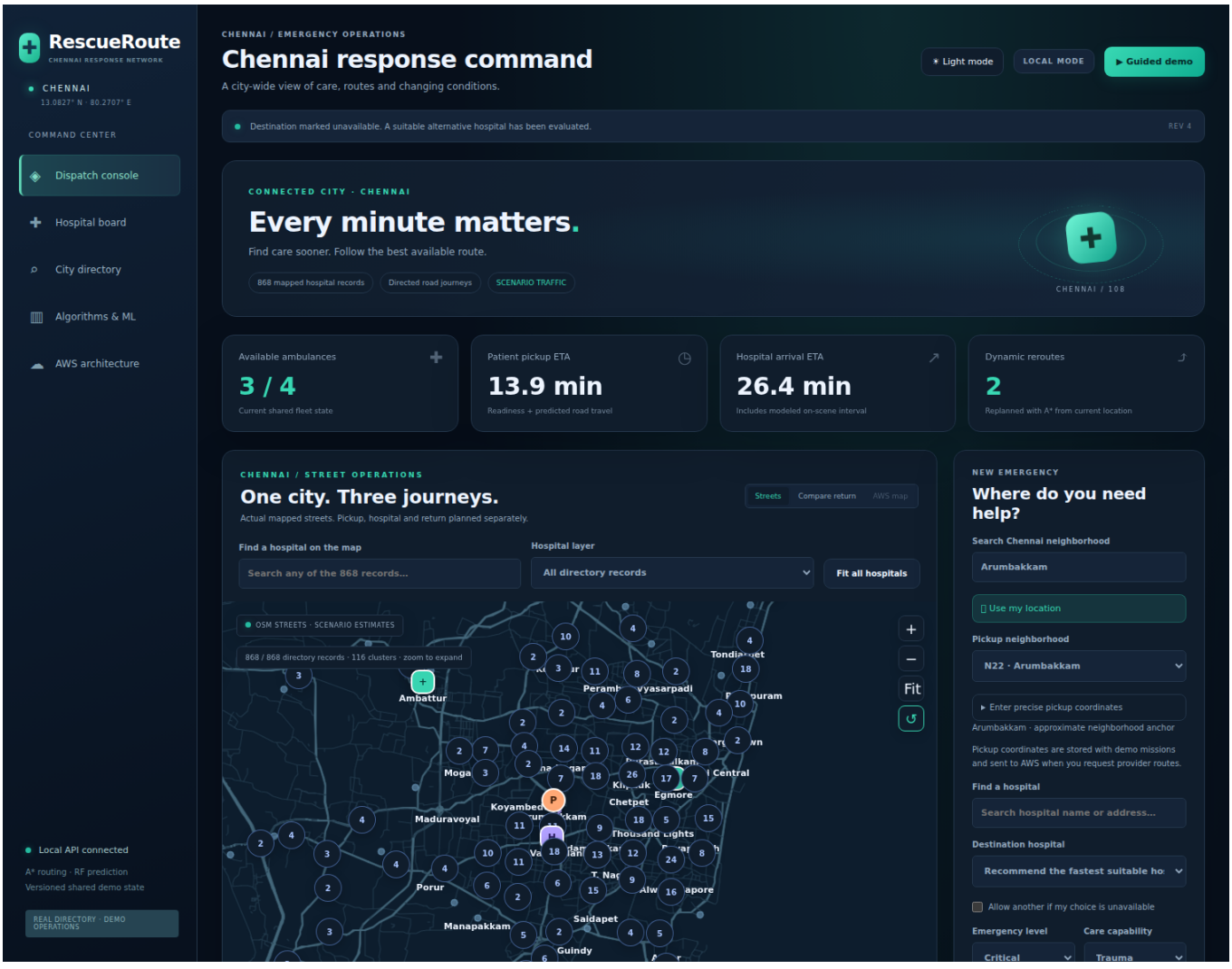
CHENNAI / AWS PROJECT REPORT

RescueRoute
Every minute matters.

Emergency ambulance route optimization with patient-selected pickup, hospital choice and five earliest destinations.

868 mapped records	5 earliest hospitals	51 passing tests
Chennai metro OSM snapshot	Travel-time ranking from pickup	Backend + offline AWS contracts



Local dashboard after a road closure and hospital reroute. Dark/light themes, animated ambulance beacons, a mapped street atlas and three independently planned directions are implemented.

Build status: Chennai application and AWS SAM package are complete and locally verified. AWS deployment was blocked because the connected browser returned Site Unavailable at the supplied console URL. No cloud resources were deployed.

01 / Chennai hospital directory

Real mapped names and coordinates make the demonstration city-specific. Operational care and bed data remain explicitly separated from map data.

Dataset detail	Bundled value
Source	OpenStreetMap amenity=hospital records, Overpass snapshot
Snapshot timestamp	8 October 2026, 11:43:05 UTC
Area	12.80-13.25 N; 80.05-80.35 E. Metro bounding box, including surrounding municipalities.
Processing	897 source elements; 875 named elements; 868 normalized records after nearby same-name merging.
Travel screen	621 name/ER-tag-screened transport candidates; heuristic, not clinical verification.
Operations	16 seeded demonstration profiles; all others unknown until operator configuration.
License	© OpenStreetMap contributors, ODbL 1.0. Attribution, raw snapshot and source IDs included.

Example hospitals available in the directory

Named hospital	Named hospital
Apollo Hospitals, Greams Road	Rajiv Gandhi Government General Hospital
MIOT International	Stanley Medical College Hospital
Kauvery Hospital, Alwarpet	Sri Ramachandra Medical Centre
Vijaya Hospital	Sundaram Medical Foundation
Billroth Hospitals, Shenoy Nagar	Government Royapettah Hospital
Kilpauk Medical College Hospital	Madras Medical Mission

These examples identify mapped institutions, not a fixed best-five list. The fastest five depend on source, travel conditions and ranking scope. No institution endorsement or verified clinical capability is implied.

Search, export and reproduce

City directory supports name/address search, profile and OSM emergency-tag filters, hospital map focus, source links, destination selection and CSV download. Missing address/phone fields remain missing.

```
python scripts/build_hospital_catalog.py
```

The command rebuilds the catalog and CSV from the bundled raw snapshot and seed mapping; it does not fetch newer data. All 868 records are retained in the directory, including possible map-tag errors or outdated entries.

Scope limit: this is not an exhaustive official hospital register. Source coordinates and emergency tags do not establish live acceptance, beds, ICU or suitability.

02 / Source, choice and earliest five

Patient-selected pickup

Choose a neighborhood, click a map point, enter latitude/longitude or request browser location. The street preview snaps to connected mapped road nodes with dotted approximate access links. The separate A* mission simulator snaps to its neighborhood anchors. Metro bounds are enforced.

Five distinct hospitals ranked by travel time

The list minimizes pickup-to-hospital predicted road minutes, adds approximate connectors and sorts ascending. It excludes ambulance arrival, on-scene time, road-risk penalties and hospital handover wait. Each card displays name, minutes, distance and capacity provenance.

Ranking scope	Meaning
Suitable demo care	Only configured care/capacity profiles that meet category, acceptance, unreserved beds and ICU requirements.
Directory travel	Screened mapped transport candidates, including unknown profiles. Travel suggestions only; no capacity or emergency suitability certification.
AWS traffic preview	If enabled after deployment, successful provider road durations replace the modeled list and display a timestamp.

Preferred destination is respected

Search any mapped record and choose a hospital. A configured preferred destination constrains dispatch even when another hospital appears earlier in the travel list. If it becomes infeasible, the mission pauses. Only explicit Allow another if my choice is unavailable permits a replacement.

An unknown hospital can be chosen for inspection or road preview, but its operator demo profile must be configured before simulated dispatch. Editing a profile does not make its capacity a verified live feed.

Traffic affects routes

Normal/rush/incident scenarios, time, rain, road congestion and closures change modeled edge weights. A* replans active missions when conditions change. A new route is chosen when it offers a better feasible objective; a different path is not guaranteed for every slider change.

Dashboard interaction and privacy

All 868 records appear through map clusters with search, zoom and hospital details. Compare return overlays outgoing/return directions; playback animates the selected road leg. Bundled street labels, dark/light themes and optional online tiles are included. Speeds/traffic remain scenario inputs.

Selected pickup coordinates are stored with demo missions/exports and sent to Amazon Location only when a provider preview is requested. Browser location is optional; denial and out-of-area locations are handled. Use demonstration points for presentations.

03 / Routing and mission lifecycle

Two objectives with separate explanations

Decision	Objective
Earliest five	Minimum pickup-to-hospital edge minutes + approximate connectors.
Ambulance assignment	Readiness + predicted journey + on-scene time + rough-road penalties + handover wait + connectors.
Arrival ETA	Predicted readiness/journey/on-scene/connector time; excludes road penalty and handover wait.

The automatic mission destination can differ from the first travel-only card because assignment considers ambulance position, waiting and road penalties. The dashboard exposes those components instead of treating every score as arrival time.

Directed street preview and separate A* mission engine

The mapped atlas contains 10,740 driving-corridor ways, 25,595 segments and 21,905 road nodes; 6,010 ways are one-way. Dijkstra separately searches ambulance-to-pickup, pickup-to-hospital and hospital-to-return. Return targets pickup or ambulance base; it is never simply reversed. Street closures and eastbound/westbound congestion are browser-local scenarios. Access connectors are approximate; turn restrictions, conditional access and barriers are not modeled.

The graph contains 36 approximate neighborhood anchors and 65 modeled undirected links. The heuristic is haversine distance multiplied by the smallest current edge-objective/geographic-distance ratio. Triangle inequality makes it conservative and consistent. Dijkstra checks weighted optimality; independent Floyd-Warshall results provide another oracle.

BFS optimizes fewest links. Shortest-distance Dijkstra optimizes kilometers. Algorithms are compared on identical endpoints/snapshots. Replanning runs A* again; D* Lite is not implemented. Correct search does not establish real-world travel prediction accuracy.

Lifecycle and reservations

Phase / event	Behavior
TO_PICKUP	Assigned ambulance advances through modeled junctions.
ON_SCENE	Modeled service interval and pickup connector travel are accounted.
TO_HOSPITAL	Route replans from current position; valid destination retained on normal GPS steps.
ARRIVED	Hospital connector time/distance accounted; reported bed/ICU capacity consumed.
COMPLETED	Ambulance released at its hospital anchor, not teleported to base.
NO_ROUTE	Mission pauses until road, destination or capacity conditions permit a route.

Active reservations reduce available hospital capacity. Earlier missions have priority during reallocation. Duplicate command IDs are protected and stale dispatch revisions rejected. SQLite immediate transactions serve local mode; DynamoDB conditional revision writes serve AWS.

The bounded four-ambulance, single-world design caps history and guards item size. It is a demonstrator, not a production multi-city dispatch database. Junction stepping and accelerated playback are simulated GPS.

04 / Synthetic ML dataset and model

Reproducible training

10,920 generated observations: 65 modeled roads x six sampled hours x 28 dates. Seed 42. Train: 1-21 January 2026 (8,190 rows). Holdout: 22-28 January (2,730). Roads recur across both periods; this tests future synthetic dates on known roads.

RandomForestRegressor: 40 trees, maximum depth 9, minimum leaf size 5. The target is a travel-time multiplier over free-flow minutes. JSON tree arrays permit standard-library inference in Lambda/local mode; twenty exported fixtures are checked against sklearn.

Feature family	Input
Road	Approximate distance_km, free-flow speed_kmh and encoded road_type
Time	Hour sine/cosine and weekday
Conditions	Synthetic/operator congestion index and rain level

Final synthetic holdout results

Predictor	MAE min	RMSE min	R-squared
Random Forest	0.624	0.916	0.995
Traffic-rule baseline	1.333	1.922	0.976
Fixed speed	11.050	14.781	-0.398

These are per-road minute errors, not full-mission arrival errors. Congestion contributes about 87.1% and rain 11.6% of impurity-based importance. Those values reflect the synthetic generation mechanism, not causal evidence about Chennai.

Run training again

```
python -m pip install -r ml/requirements.txt
python ml/train.py
```

Training regenerates the CSV, forest JSON, report and inference fixtures. Merely replacing the CSV does not retrain the supplied model. The dashboard supports an ML-off traffic-rule ablation and algorithm comparisons.

Limitations and real-data extension

No real Chennai traffic training data or observed ambulance trips are bundled. Amazon Location preview results are not used to train this forest. High holdout R-squared is expected from a shared synthetic generator; no hyperparameter search used the holdout.

A real extension needs authorized de-identified measurements, documented units/provenance, chronological validation/test splits, simple baselines, calibration and route/mission arrival error evaluation under drift and outages. The supplied metrics are not a claim of real-world accuracy.

05 / AWS services and deployment

Service	Implemented responsibility
Cognito / API Gateway	Operator login; HTTP API JWT validation, CORS and throttling.
Lambda	A* missions, directed street preview, JSON ML inference, exports and Location calls.
DynamoDB	Shared demo state with consistent reads and conditional revisions.
S3 / CloudFront	Private frontend with OAC; private model/data/catalog and report exports.
CloudWatch / IAM	Seven-day logs, latency/error metrics, dispatch alarm and scoped permissions.
Amazon Location	Optional map-only restricted browser key; server-side matrix and route calls.

Deploy from an AWS-configured computer or CloudShell

```
aws configure
aws sts get-caller-identity
sam build
sam deploy --guided --stack-name rescueroute-chennai --region eu-north-1
python -m pip install -r requirements-aws.txt
python scripts/publish_frontend.py --stack rescueroute-chennai
python scripts/setup_user.py --stack rescueroute-chennai --username dispatcher
```

Scripts default to eu-north-1. For another region pass --region consistently. An activated account, AWS CLI/SAM, Python 3.12+ and provisioning permissions are required. Confirm IAM creation in SAM. Start with EnableLocation=false. Operator setup prompts privately for a password.

Enable provider roads / traffic

```
sam deploy --stack-name rescueroute-chennai --region eu-north-1
--parameter-overrides EnableLocation=true
python scripts/publish_frontend.py --stack rescueroute-chennai
```

Enter the sam deploy command above on one line. Republish after enabling Location. Real roads & traffic calls CalculateRouteMatrix with actual coordinates, DepartNow, Car, FastestRoute and UseTrafficData, then CalculateRoutes for geometry and up to two alternatives. Traffic depends on provider coverage.

Directory scope can request hundreds of paid matrix pairs per click; configured-care scope is smaller. The response shows scope, timestamp and pairs. Provider preview is independent of A* missions and does not certify capacity or implement emergency driving exemptions. Disabled/failed provider requests never produce fake live traffic.

Deployment evidence is still required. No cloud stack was deployed from this session because the console returned Site Unavailable. Bundled screenshots and SDK stubs are local evidence only. There is no SageMaker or SMS/email integration. See docs/AWS_DEPLOYMENT.md for cloud checks, troubleshooting and removal.

06 / Verification and demonstration

Completed checks

Check	Result
Backend / SDK	51 tests passed with boto3: 47 core + four offline SDK adapter tests.
Graph optimality	3,888 A*/Dijkstra endpoint comparisons, heuristic consistency and independent Floyd-Warshall checks.
Chennai behavior	868 records/CSV, five minimum-time distinct hospitals, bounds, unknown profiles, preferred destination and fallback permission.
Browser workflow	Road/hospital reroutes, shared operator state, draft preservation, source coordinates, selected MIOT, mocked geolocation, directory ranking, completion and export.
Visual / responsive	Dark/light, three road legs, return comparison, 868 map records, street closures, playback and 390px mobile without document overflow; no workflow JavaScript errors.
Infrastructure / provenance	cfn-lint validated; SDK request contracts stubbed; normalized catalog rebuilt identically.

Run the checks yourself

```
python -m unittest discover -s tests -v
npm install
npx playwright install chromium
npm run test:browser
```

Without boto3, the four SDK tests skip. The browser test starts a fresh temporary database on port 8022 and regenerates screenshots/results. SDK stubs make no provider calls; geolocation is mocked. Cloud permissions, provider coverage, real GPS precision and service availability require actual deployment checks.

Eight-minute presentation sequence

Show Chennai data and source selection; inspect five hospitals; choose MIOT; run Guided demo; inject a road closure and hospital unavailability; advance to arrival and completion; compare algorithms/ML; compare the independently planned outgoing/return routes, then show AWS service responsibilities and export a report. The tested automatic destination changes H15 Apollo First Med to H08 Vijaya after the injected events.

Submission and safety boundaries

Include your own SAM outputs, CloudFront/Cognito sign-in, DynamoDB revision, S3 report, CloudWatch metrics and optional Location response to demonstrate actual AWS use. No production load/security audit, real-data evaluation or clinical validation was performed. Do not use this educational simulation for emergency dispatch.

Primary sources and bundled references

OpenStreetMap: openstreetmap.org/copyright
Chennai district: chennai.nic.in/public-utility-category/hospitals/
AWS: docs.aws.amazon.com/location/latest/APIReference/
Bundled: README; docs/DEMO_GUIDE, AWS_DEPLOYMENT, DATA_CARD, VERIFICATION and STREET_MAP_UPDATE; backend/city/LICENSE.txt.